

BLACKSMITH

MASTER PRODUCTION GDD

강화의 긴장감과 한 작품의 생애를 연결하는 Android 세로형 대장간 게임

기준 main	b4a3b01e247b4c326890822d78ca5932fe6d4ff6
문서 생성일	2026-08-28 KST
현재 상태	Implementation and Review · Human/Android/UX evidence NOT_RUN

이 PDF는 사람용 제작 기획서입니다. 현재 정본, 실제 Godot 구현, 검증 증거를 구분해 읽을 수 있게 작성했습니다.
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읽는 순서

무슨 게임인가 → 플레이어가 무엇을 판단하는가 → 강화·내구·수리·고객 생애가 어떻게 연결되는가 → Godot에서 어떤 책임으로 구현되는가 → 무엇이 구현·검증됐고 무엇이 남았는가의 순서입니다.

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Document role: machine-searchable current-canon planning and implementation contract. It is not a replacement for executable code, JSON owners, tests, or runtime evidence.

Source snapshot: `main / b4a3b01e247b4c326890822d78ca5932fe6d4ff6` / 2026-08-28 KST.

Generated scope: exactly one AI specification paired with `exports/blacksmith_MASTER_PRODUCTION_GDD_20260828.pdf`.
No Notion write, new raster, runtime asset, Scene, data, or GDScript change belongs to this document batch.

00. CANON SNAPSHOT

Field	Current value
Project	Blacksmith - Android portrait Godot crafting game
Current product mode	CURRENT_CANON_MVP_ACTIVE_BY_USER_DECLARATION_20260826 / IMPLEMENTATION_AND_REVIEW
Primary core	Enhancement tension plus DDD; the recurring player question is STOP OR PUSH.
Differentiator	The same item UID carries ownership, damage, repair, handoff, actual use, meaningful events, and chronicle through a customer-facing life.
Current visual direction	ILLUSTRATED_WORKSHOP_BOOK / USER_APPROVED_DIRECTION; warm hand-drawn workshop book, paper/leather/iron/wood material language, readable modern controls.
Canonical current owners	BLACKSMITH_CORE_SIMPLIFICATION_CANON_20260825.md; Decisions 28-32 and 34 JSON/decision owners listed below.
Open PR boundary	PR #196 is OPEN_DRAFT_READ_ONLY_DO_NOT_TAKE_OVER; it is not merged truth.
Evidence ceiling	Automated contracts and selected GUT/CI evidence exist. Godot client rendering, Android, accessibility, performance, Human usability, player experience, and release readiness remain NOT_RUN.

Status vocabulary

DOCUMENTED means this specification records the claim. CONFIRMED means it is approved by the current source owner. IMPLEMENTED means a current repository path exists, not that it matches all current canon.

AUTOMATED_TEST_PASS needs named passed test evidence. RUNTIME_VERIFIED, UX_VERIFIED, and RELEASE_READY require their own direct evidence and must never be inferred from earlier states.

01. SOURCE REGISTRY

ID	Source / use	Status
SRC-CAN-01	AGENTS.md	Current authority order, protected scope, test and evidence rules.
SRC-CAN-02	docs/operations/BS-OPS-20260825-08_SESSION_HANDOFF_CORE_SIMPLIFICATION.md	Cold-start locator and accepted frontier.
SRC-CAN-03	docs/planning/BLACKSMITH_CORE_SIMPLIFICATION_CANON_20260825.md	Current player promise, enhancement-first loop and protected rules.
SRC-CAN-04	docs/planning/BLACKSMITH_DAMAGE_PROBABILITY_CURVE_20260826.json	Decision28 exact conditional damage curve and Decision32 outcome/display ownership.
SRC-CAN-05	docs/planning/BLACKSMITH_DURABILITY_REPAIR_MODEL_20260826.json	Decision29 visible durability authority and repair/scar model.
SRC-CAN-06	docs/planning/BLACKSMITH_REPAIR_ECONOMY_REBASE_20260826.json	Decision31 test-only repair-cost curve.
SRC-CAN-07	docs/planning/BLACKSMITH_CUSTOMER_WORLD_EVENT_DAMAGE_POLICY_20260826.json	Decision30 actual-use event eligibility and damage composition.
SRC-CAN-08	docs/decisions/BS-ENHANCE-20260828-34_WEAPON_KEYWORD_OWNERSHIP.md	Grade / Tag / Event taxonomy and +9→+10 keyword ownership.
SRC-CAN-09	docs/planning/BLACKSMITH_ACTUAL_GAME_IMAGE_CONSUMER_GATE_20260826.json and assets/ASSET_MANIFEST.json	Visual consumer, asset provenance, and validation state.
SRC-IMP-01	project.godot, scenes/vertical_slice/, scripts/vertical_slice/	Current vertical-slice implementation reality.
SRC-IMP-02	data/vertical_slice/, data/crafting/, tests/gut/unit/vertical_slice/	Current data and automated-test reality.
SRC-HIS-01	[기획서]00_프로젝트_허브/ACTIVE_CONTEXT.md, R2/R3 Game Bibles, legacy POC scripts	Historical / compatibility evidence only where it conflicts with SRC-CAN owners.
SRC-BMK-01	Diablo IV itemization update, accessed 2026-08-28	Official comparative evidence for choosing an explicit single item-affix result and separate quality path.
SRC-BMK-02	Last Epoch Support: Forging Potential, accessed 2026-08-28	Official support index confirms a finite item-forging resource as a distinct craft boundary; no numeric import.
SRC-BMK-03	Shop Titans on Google Play, accessed 2026-08-28	Official market page for crafting, visitor sale, shop growth, and mobile production-loop comparison.
SRC-BMK-04	Moonlighter on Steam, accessed 2026-08-28	Official product page for shopkeeper/day-life, price feedback, crafting/enchanting and villager relation comparison.
SRC-BMK-05	Potion Permit help, accessed 2026-08-28	Official example of customer need → diagnosis → gather → make → result loop.
SRC-PLT-01	Godot Android export, accessed 2026-08-28	Official Android export and signing/AAB requirements.
SRC-PLT-02	Android quality checklist and Android vitals, accessed 2026-08-28	Platform quality, continuity, crash/ANR, memory and bitmap-risk evidence.

Source and conflict register

Conflict ID	Competing claims	Actual implementation reality	Current disposition
DEC-DRIFT-01	Historical multiple precision milestones and catalyst evolution vs current +9→+10 only.	Old data and legacy code contain older milestone vocabulary.	SUPERSEDED: current canon allows exactly one Precision Enhancement at +9→+10.
DEC-DRIFT-02	Historical CURRENT/MAX bands and overhaul formulas vs Decision29 CURRENT/MAX/BASE_MAX.	V2 field names may resemble current ones but do not establish semantic compliance.	SUPERSEDED: visible numeric authority and derived effective state are Decision29 only.
DEC-DRIFT-03	Catalyst-only Tag identity vs latest direct user rule.	No approved Tag write or player presentation exists.	CONFIRMED: Tag identity resolves from catalyst lineage plus selected Precision method; see SYS-PRE-002.
DEC-DRIFT-04	Historical visual-GDD image production vs current consumer gate.	Three approved runtime asset families are dynamically bound; client visual review is not run.	CONFIRMED: no new raster is presumed from a document gap; consumer and rights status are required.
DEC-DRIFT-05	PR #196 proposed precision visitor context vs main.	Draft adds only a test file and has one failed authority check.	OPEN_DRAFT_READ_ONLY; do not merge, cite, or treat as implementation.

02. CURRENT PROJECT STATE

Work 5-stage readback

Work stage	State	Evidence / boundary
1. Intent and canon	CONFIRMED	Current owner files and Decisions 25-34.
2. System contract	PARTIAL	Enhancement, durability, repair, event, and keyword boundaries are current; Tag content matrix and empty-lineage policy are unresolved.
3. Representative slice plan	CONFIRMED	Slice B: individual +0→+10, optional +11→+15 risk, handoff, brief return beat, same-UID actual-use result.
4. Machine implementation	PARTIAL / IMPLEMENTATION_DRIFT_RISK	Vertical-slice classes, scenes, save envelope, GUT contracts and assets exist; exact current-canon parity is not asserted.
5. Player and production proof	NOT_RUN	No Godot client, Android, accessibility, performance, or human/player test evidence.

Actual playable-surface inventory

UI ID	Surface	Existing locator	Implementation state
UI-MAIN-001	Main Menu	res://scenes/vertical_slice/main_menu.tscn, VSMainMenu	IMPLEMENTED; new/continue/settings stub.
UI-FORGE-001	First Forge	scripts/ui/forging_screen.gd and first-forge flow	IMPLEMENTED historical/V2 surface; current flow parity requires review.
UI-WORK-001	Workshop	res://scenes/vertical_slice/screens/vs_workshop_screen.tscn, VSWorkshopScreen	IMPLEMENTED for numeric durability/repair surface; current canon parity not claimed.
UI-RESULT-001	Customer Result	res://scenes/vertical_slice/screens/vs_customer_result_screen.tscn, VSCustomerResultScreen	IMPLEMENTED presentation surface; current event/content contract parity requires review.
UI-PREC-001	+9→+10 Precision	Only locator / future consumer queue	NOT_IMPLEMENTED as an approved current screen.
UI-CHR-001	Item Chronicle	Only locator / future consumer queue	NOT_IMPLEMENTED.

03. CONFIRMED DECISIONS

Decision ID	Confirmed rule	State
DEC-ENH-25	Ordinary enhancement success is always +1; only +9→+10 is Precision Enhancement.	CONFIRMED
DEC-DMG-28	Target +11/+30/+60/+90/+100 conditional-on-failure damage anchors are 5/6/7/8/10%; exact piecewise-linear interpolation.	CONFIRMED / TEST_BUDGET
DEC-REP-29	CURRENT/MAX/BASE_MAX is the only visible durability authority; state is derived from the worse ratio.	CONFIRMED / TEST_BUDGET
DEC-REP-31	One repair job after actual damage; current initial curve is a sensitivity-tested budget, not final economy.	CONFIRMED / NOT_FINAL_BALANCE
DEC-DMG-30	Handoff itself causes no damage; actual use is required; one roll per event per UID; world event never directly damages MAX.	CONFIRMED
DEC-ENH-32	A failed enhancement resolves to exactly FAILED_HOLD or FAILED_DAMAGE; no downgrade or separate critical.	CONFIRMED
DEC-CHR-27	Only meaningful item events enter player Chronicle, never routine dated attempts.	CONFIRMED
DEC-ENH-34	Weapon keywords are Grade / Tag / Event. +10 success creates exactly one weapon-owned Tag in CATALYST_AFFIX; its identity is catalyst lineage plus selected Precision method.	CONFIRMED
DEC-ART-03/04	Illustrated Workshop Book direction; generated imagery needs actual runtime consumer and is not automatically a final asset.	CONFIRMED

04. DESIGN PILLARS

1. SYS-ENH-001 - Push matters because stopping is valid. Every preview must make the next attempt's gain, cost, risk, and meaningful loss legible enough that stopping is a player decision, not a hidden-information trap.
2. SYS-LIF-005 - One weapon remembers. The UID connects forge, risk, repair, owner, actual use, outcome, and chronicle, so an item is not disposable spreadsheet output.
3. SYS-CUS-006 - Customers provide purpose, not a combat minigame. Player judgment is about which work to make and risk, while the world result returns causal evidence rather than an opaque score.
4. SYS-FDB-009 - Five-level rises mark craft progress. Every fifth level must visibly raise craft feeling; +5 is feedback only and +10 remains the sole special boundary.
5. UI-UX-001 - Android portrait readability first. Exact current/max/base-max, reason text, state labels, and touch targets carry the meaning; art cannot replace them.

05. PLAYER EXPERIENCE CONTRACT

Promise chain

Customer purpose → make one named UID work → read next enhancement preview → stop or push → gain +1 or receive a clear hold/damage consequence → hand off the same work → receive a delayed actual-use result with 2-4 causes → repair, continue, retire, or begin a new work.

The target first memory is not "I clicked a forge many times." It is "I chose to protect or push *this* weapon, and the world later showed what happened to it." The emotion curve is anticipation, relief/pride, tension, responsibility, and curiosity about the returned life result.

First 5 / 15 / 30 minute contract

UX ID	Time	Player learns	Evidence state
UX-ONB-001	First 5 min	Make a first work and see ordinary +1 feedback; +1/+2 learn and +3~+9 build.	DOCUMENTED; pacing UX_VERIFIED: NOT_RUN.
UX-ONB-002	First 15 min	+9→+10 is special: the successful work gains one Tag, checkpoint/break-even meaning, then +11 previews first damage-eligible risk.	CONFIRMED; precision screen/data content NOT_IMPLEMENTED.
UX-ONB-003	First 30 min	Handoff produces a non-economic return beat and an actual same-UID result, causing the next repair/enhancement/new-work choice.	DOCUMENTED / PARTIAL; runtime/UX proof NOT_RUN.

06. CORE / SESSION / META LOOP

CORE: Forge UID → preview next +1 → STOP or PUSH → clear result → next decision

SESSION: customer purpose → make/upgrade a work → handoff → delayed actual-use result → choose repair/continue/new work

META: build a portfolio of remembered UIDs → unlock later title/content only through approved systems → deepen customer/world consequences

+10 is a safe-no-damage boundary and a single identity moment. +11 is the first salient voluntary risk. +100 is terminal enhancement completion, not a reset/prestige loop. The 6-8 minute session figure is a playtest hypothesis, never a hard time limit.

07. SYSTEM REGISTRY

System ID	Name	Current owner	Status
SYS-ENH-001	Ordinary enhancement and STOP/PUSH	Core simplification canon	CONFIRMED; V2 implementation PARTIAL
SYS-PRE-002	+9→+10 Precision and weapon Tag	Decision34 / unified contract	CONFIRMED; Tag content/write/UI NOT_IMPLEMENTED
SYS-DUR-003	Durability and damage	Decision28/29	CONFIRMED / TEST_BUDGET; V2 parity NOT_CONFIRMED
SYS-REP-004	Repair and structural scar	Decision29/31	CONFIRMED / NOT_FINAL_BALANCE; V2 parity NOT_CONFIRMED
SYS-LIF-005	UID, ownership, destruction, archive	Core canon / save contract	CONFIRMED; V2 implementation PARTIAL
SYS-CUS-006	Customer/world actual-use result	Decision30 / content contract	CONFIRMED; content breadth PARTIAL
SYS-CHR-007	Meaningful Chronicle	Decision27	CONFIRMED; player UI NOT_IMPLEMENTED
SYS-ECO-008	Enhancement/repair economy	Decision31 and existing budgets	PARTIAL / TEST_IN_PLAY
SYS-FDB-009	5-level craft-rise feedback	Phase1 unified contract	CONFIRMED; visual/runtime proof NOT_RUN

08. SYSTEM SPECIFICATIONS

SYS-ENH-001 - Ordinary enhancement and STOP/PUSH

Player contract. The player sees a next target, cost, success possibility, failure consequence, and relevant durability state, then chooses STOP or PUSH. The immediate reward is +1; the long-term reward is a more capable, more meaningful work. The system exists so risk is a comprehensible judgment, not an optimization-only meter.

Contract area	Current rule
Entry	A non-destroyed item UID and a legal next target.
Input	Explicit enhancement attempt; no auto-hidden outcome.
Success	Exactly SUCCESS_LEVEL_DELTA = +1.
Failure	FAILED_HOLD or FAILED_DAMAGE only. No level downgrade and no critical result.
Damage gate	Target $\leq$ +10: no enhancement damage. Target $\geq$ +11: only after failure, with Decision28 $\times$ Decision29 calculation.
Exit / recovery	Player may stop at any legal point; actual damage can unlock one repair job; destroyed item is not repairable.
Required feedback	Target, exact displayed final probabilities rounded to one decimal, resource cost, durable state, result, and next legal action.
Implementation locators	scripts/vertical_slice/resolvers/vs_enhancement_resolver.gd, scripts/vertical_slice/services/vs_enhancement_action_service.gd, scripts/vertical_slice/ui/vs_workshop_screen.gd. Existing paths are implementation reality, not parity certification.
Acceptance	Same supplied deterministic rolls always resolve to one legal result; +10 or lower cannot damage; +11+ never produces downgrade/critical; save is atomic.

SYS-PRE-002 - +9→+10 Precision and Tag resolution

Player contract. At the only precision boundary, the player chooses a method in a disclosed catalyst lineage context. Success gives one weapon-owned Tag that names a meaningful identity of that weapon; the method also affects only the weapon's stats, durability, and Tag-resolution context. It does not create player keywords, titles, Grade, Event, or a fourth slot.

Field / rule	Contract
Trigger	Only a successful +9→+10 enhancement.
Storage	Exactly one CATALYST_AFFIX / Tag on that weapon UID.
Resolution	tag_id = resolve(catalyst_lineage, precision_method); the stored outcome is a single Tag, even if its player-facing copy expresses both inputs.
Isolation	GRADE_AFFIX and CHRONICLE_AFFIX cannot be written, changed, or selected by the method.
Unresolved	First lineage $\times$ method $\times$ Tag content matrix, exact copy, empty-lineage behavior, UI entry/exit, and save/write implementation.
Prohibited	Placeholder Tag, invented keyword, player title, event keyword, universal damage protection, fourth affix slot.
Required future tests	One valid combination returns one known Tag; invalid / absent lineage follows an explicitly approved policy; failure creates no Tag; Grade/Event remain byte-for-byte unchanged.

SYS-DUR-003 - Durability and damage

Player contract. The player can understand whether a work is normal, minor, major, or destroyed from visible numbers. They feel a worsening item life without hidden double penalties.

```
0 <= CURRENT <= MAX <= BASE_MAX
current_ratio = CURRENT / MAX
structural_ratio = MAX / BASE_MAX
effective_ratio = min(current_ratio, structural_ratio)
NORMAL = 1.00; MINOR = (0.50, 1.00); MAJOR = (0, 0.50); DESTROYED = CURRENT == 0
```

State	Success modifier	New ordinary effect	Failure-damage multiplier
NORMAL	0pp	×1.00	×1.00
MINOR	-3pp	×0.90	×1.25
MAJOR	-7pp	×0.75	×1.75

These values are TEMP_TEST_BUDGET, not release balance. A hard guarantee remains truly 100%. The UI must display a final attempt chance rounded to one decimal but resolve exact values. One actual event causes at most one damage event for the UID.

SYS-REP-004 - Repair and scar

Player contract. A real loss opens one concrete chance to restore an item, but restoration may leave an understandable structural mark. It should motivate another decision, not add a routine tax.

Rule	Contract
Eligibility	0 < CURRENT < MAX and REPAIR_JOB_AVAILABLE; destroyed and full-durability repair are forbidden.
Job consumption	Consume repair job at repair start.
Quality budget	Excellent 20% → MAX 100%; Standard 60% → MAX 75%; Poor 20% → MAX 50%.
Scar	Pre-repair effective state × enhancement band determines a temporary MAX -1 trigger chance; skip scar with no positive CURRENT gain, no reroll.
Initial cost	ceil(R_BAND * (0.05 + 0.65 * ((MAX-CURRENT)/BASE_MAX))) + 1 reinforcement. Sensitivity b=.50/.65/.80 remains required.
Future acceptance	Quote and apply share one resolver; no repair job can be duplicated with reload/retry; one repair cannot cause zero positive CURRENT recovery.

SYS-LIF-005 - UID, ownership, destruction, archive

An item UID is born once and retains birth facts, enhancement, meaningful change ledger, ownership/handoff, actual-use results, repair/scar, destruction reason, and predecessor relation. A destroyed physical item cannot be revived, but its history persists. A successor is a new UID and inherits no power or chronicle data.

DAT-SAVE-001 uses V3 save and item schemas in data/vertical_slice/vertical_slice_preset.json: deterministic birth/run seeds, append-only contiguous ledger, resolved events, temp-flush-rename backup, V2 migration on read, V1 fail-closed.

SYS-CUS-006 - Customer/world actual-use result

Customers make the work meaningful before and after the forge; they do not turn Blacksmith into direct exploration or combat. A valid event requires actual use of the same UID. Purchase or handoff alone never damages the item. Mission outcome and damage are independent axes; an event has at most one damage roll for a UID.

CNT-ADVENTURER-01 / NADIA_VENN is the starter representative: SURVIVAL_AND_RECOVERY with EXPEDITION_RETURN_STATE, RECOVERY_STATE, and ITEM_UID_LIFECYCLE_STATE. A result needs 2-4 causal reasons and one primary next action. Generic Tag cannot reduce a DIRECT event; explicitly causal relevant item property may lower a probabilistic profile by at most one step.

SYS-CHR-007 - Meaningful Chronicle

Chronicle is a player-facing explanation layer for creation, Tag/keyword, actual damage, MAX-scar repair, handoff, world result, destruction, and comparable meaningful events. Routine enhancement attempts and date-spam are prohibited. UI-CHR-001 is not implemented; therefore this contract does not pretend that a current player can inspect the history surface.

SYS-ECO-008 and SYS-FDB-009

Economy supports enhancement choices, not passive grind. Existing attempt cost and reinforcement supply are USER_APPROVED_TEST_BUDGET / NOT_FINAL_PRODUCT_BALANCE; legacy expected-cost tables require recalculation after current damage/repair play evidence. Every 5 levels must provide a craft-rise feedback beat; +5 is feedback only, while +10 remains the sole precision event. Required production proof is a representative animation/VFX/SFX and readability test, currently NOT_RUN.

09. CONTENT REGISTRY

Content ID	Customer / content	Purpose	Current status
CNT-ADVENTURER-01	Nadia Venn	Starter same-UID expedition survival/recovery result.	DOCUMENTED / PARTIAL IMPLEMENTATION
CNT-ADVENTURER-02	Toren March	Journey arrival, route exposure and item lifecycle.	DOCUMENTED
CNT-SOLDIER-01	Marek Olden	Small-lot standard order and mission result.	DOCUMENTED
CNT-COLLECTOR-01	Ersa Roen	Exhibition reception, thesis fit, public item legacy.	DOCUMENTED
CNT-GLADIATOR-01	Cassia Bellan	Arena result and equipment contribution, not direct combat.	DOCUMENTED
CNT-NOBLE-01	Ceremonial Noble	Ceremony readiness and heirloom treatment.	DOCUMENTED
CNT-SOLDIER-02	Liana Berg	Commander mission and field legacy.	DOCUMENTED
CNT-COLLECTOR-02	Sedric Vael	Archive accession, provenance, custody legacy.	DOCUMENTED
CNT-GLADIATOR-02	Kyle Varen	Veteran return / keep-in-service vs retire-replace.	HISTORICAL_REUSE_REFERENCE / DOCUMENTED

10. CONTENT SPECIFICATIONS

CNT-ADVENTURER-01 - Nadia Venn starter slice

Item	Contract
Entry	First forged same UID is available for handoff; Nadia's public identity is ELITE, but exact numeric capability profile is explicitly SEPARATE_CANON_SOURCE_REQUIRED.
Player judgment	Read disclosed purpose, constraint, and known situation; decide whether the work is ready at +10 or worth a voluntary +11 risk.
Outcome	Handoff → one brief return beat → actual expedition use → CONTENT_RESULT_V1 record with exactly 2-4 causal reasons and one next action.
Damage boundary	No damage at handoff. Eligible actual use may produce at most one UID damage event under SYS-CUS-006 and SYS-DUR-003.
Prohibited	Inventing Nadia's capability values, best badge, universal fit score, direct expedition play, automatic Tag/Chronicle farming.
Data / implementation locators	data/vertical_slice/customers/nadia_venn.json; data/vertical_slice/content_result_contract.json; VSContentResultRecord, VSCustomerActualUseActionService, VSCustomerWorldEventResolver.
Acceptance	Result rejects an unrelated UID, forbids a score/reward field, has valid axes/reasons/next action, and does not reroll on load.

Content production template

Each additional content row must specify: customer goal and visible constraint; trigger and exit; required work UID; allowed result axes; 2-4 causal reasons; one next action; event damage eligibility; no-direct-combat boundaries; required copy/UI/audio/visual consumers; data owner; save impact; acceptance tests; and DOCUMENTED / IMPLEMENTED / RUNTIME_VERIFIED / UX_VERIFIED evidence separately. No content may use a hidden total score, automatic artistry growth, automatic Chronicle grants, or fixed universal result dates.

11. UI/UX AND INPUT CONTRACT

UI / UX ID	Goal and required information	Input / feedback	State
UI-MAIN-001	Start/continue without misrepresenting save state.	New Game, Continue, Settings stub; visible save status.	IMPLEMENTED / UX_NOT_RUN
UI-FORGE-001	Establish first work and ordinary forge feedback.	Touch forge; clear progress/result.	PARTIAL IMPLEMENTATION
UI-WORK-001	Compare exact durability and repair availability before action.	Enhancement / repair actions; disabled/error states communicate why.	IMPLEMENTED / CURRENT_PARITY_NOT_CONFIRMED
UI-PREC-001	Make the lineage + method choice and one resulting Tag intelligible.	Explicit lineage/method selection, preview, confirm/cancel; success/failure result.	NOT_IMPLEMENTED
UI-RESULT-001	Explain returned customer outcome and next action.	Continue/back action; 2-4 reasons, damage state, next action.	IMPLEMENTED / UX_NOT_RUN
UI-CHR-001	Browse only meaningful UID events.	Open detail, filter/read, close.	NOT_IMPLEMENTED

Accessibility and touch rules. Portrait target is 720×1280 logical viewport with orientation locked portrait. Status must be text-plus-icon/shape, never color alone. A destructive or irreversible-looking push requires an explicit confirmation only when that interaction has been approved; duplicate taps must be prevented at resolver/service level. Every disabled control states why, error recovery retains the selected UID, and target touch size is at least the existing project 48dp policy. Current Android, screen-reader, contrast, large-text, motion, and device safe-area evidence is NOT_RUN.

14. Visual Bible · Current approved references

아래 자료는 현재 저장소에 승인 기록이 있는 시각 참고 또는 실제 런타임 소비 자산입니다. 이미지는 실제 화면 검증을 뜻하지 않으며, 새 이미지를 이 문서 작업에서 생성하지 않았습니다.



AST-REF-004 · 제작용 Illustrated Workshop Book reference · runtime consumer 없음



AST-WKS-001 · Workshop/Main Menu 배경 · client/Android visual QA NOT_RUN



AST-DUR-002 · NORMAL/MINOR/MAJOR/DESTROYED atlas · 수치 내구도는 계속 권위

12. VISUAL ASSET CONSUMER MATRIX

AST ID	Asset	Actual consumer	Status and boundary
AST-WKS-001	ASSET-WORKSHOP-BACKGROUND-V2	VSWorkshopScreen; VSMainMenu dynamic override	PROJECT_ASSET_APPROVED, static/dynamic binding recorded; Godot client/Android/accessibility/human visual validation NOT_RUN; release rights pending.
AST-DUR-002	ASSET-WORKPIECE-DURABILITY-STATE-ATLAS-V1	WorkshopLayout/WorkpieceDurabilityHero	Four state cells; native numeric authority remains primary; dynamic binding recorded, visual validation NOT_RUN.
AST-FORGE-003	ASSET-FIRST-FORGE-BACKGROUND-V1	ForgingScreen / FirstForgeIllustratedBackground	Project-approved runtime family; readability veil; client visual validation NOT_RUN.
AST-REF-004	ASSET-ILLUSTRATED-WORKSHOP-BOOK-REFERENCE-V1	None by design	Approved production reference, never a runtime/marketing substitute.
AST-MKT-005	Key art / app icon masters	None by design	Release drafts, not platform-ready; rights/terms/safe-zone work required.
AST-PREC-006	Precision visual treatment	No exact consumer slot	DEFERRED; do not generate until UI-PREC-001 defines its consumer, states, aspect, and fallback.
AST-CHR-007	Chronicle visual treatment	No exact consumer slot	DEFERRED; do not generate until UI-CHR-001 exists.

All imagery follows the same grammar: warm illustrated workshop book; hand-drawn paper/leather/iron/wood; calm warm light; readable native Godot UI above it. Fake generated gameplay screenshots, image-only system truth, copied game identity, and promotion of reference art to runtime are prohibited. Current asset rights are RELEASE_BLOCKED_UNVERIFIED until terms and distribution rights are evidenced.

13. AUDIO CONSUMER MATRIX

No approved current music/SFX/voice asset manifest or runtime audio consumer is present in the fresh-read scope.
Therefore:

AUD ID	Required future consumer	Required feedback	State
AUD-FORGE-001	Ordinary enhancement	anticipation, hit, success, hold, damage	NOT_IMPLEMENTED
AUD-PREC-002	+9→+10 precision	choice lock, resolve, Tag reveal	NOT_IMPLEMENTED
AUD-REP-003	Repair	quote accepted, quality, structural scar	NOT_IMPLEMENTED
AUD-RES-004	Customer result	return, outcome polarity, next-action cue	NOT_IMPLEMENTED

Audio must not be invented as an approved asset. Each future row needs source/provenance, player function, platform mix/readability, runtime path, and state-specific trigger before production.

14. TECHNICAL ARCHITECTURE

MainMenu (VSMainMenu)
 → run_initializer / first forge completion
 → BlacksmithApp (VSApp)
 → WorkshopScreen (VWorkshopScreen)
 → VSEnhancementActionService → VSEnhancementResolver
 → VWorkshopMaintenanceService → VSRepairResolver
 → VSSaveService / VSSaveEnvelope
 → CustomerResultScreen (VSCustomerResultScreen)
 → VSCustomerActualUseActionService → VSCustomerWorldEventResolver

Layer	Current path(s)	Responsibility
Domain	scripts/vertical_slice/domain/vs_item.gd, vs_save_envelope.gd, vs_ledger_entry.gd, vs_content_result_record.gd	UID facts, durability/item state, serialized envelope, immutable result record.
Resolver	scripts/vertical_slice/resolvers/	Pure-ish legality, damage, repair, precision, customer-context/event outcome calculations.
Service	scripts/vertical_slice/services/	Coordinates action, ledger append, save, and state changes.
UI	scripts/vertical_slice/ui/	Presents state and translates player input; must not become numeric source of truth.
Data	data/vertical_slice/, data/crafting/	Versioned preset, customer and content contracts, current/historical data fixtures.
Test	tests/gut/unit/vertical_slice/, Python contracts	GUT unit/integration and document/canon regression contracts.

Existing architecture is a reusable implementation baseline, not permission to import legacy semantics into current features. .tscn, resources, and project settings require the project-authoring provenance rules; this documentation task modifies none.

15. DATA CONTRACTS

DAT ID	Owner / essential fields	Current contract
DAT-ITEM-001	VSItem	UID, level, birth facts, CURRENT/MAX/BASE_MAX, lifecycle state, no invented Tag matrix.
DAT-SAVE-001	VSSaveEnvelope	V3; deterministic seeds, resolved events, append-only ledger, backup recovery; V2 migrate on read, V1 fail closed.
DAT-RESULT-001	CONTENT_RESULT_V1	event ID, source decision, content/customer, day, item refs, result axes, 2-4 causal reasons, one next action; score/total/reward probability prohibited.
DAT-PREC-001	Future current Tag resolution	catalyst_lineage_id, precision_method_id, tag_id; exact enum content and empty-lineage behavior USER_DECISION_REQUIRED.
DAT-DUR-001	Decision29 model	immutable BASE_MAX; legal CURRENT/MAX ranges; derived effective state is never separately persisted as hidden authority.
DAT-EVENT-001	Decision30 event packet	actual-use flag, item UID, profile/cause, resolved-once identifier; direct/max-damage constraints.

16. SCENE MAP

Scene ID	Path	Entry / exit	State
SCN-MAIN-001	res://scenes/vertical_slice/main_menu.tscn	Project entry → new/continue first forge/workshop	IMPLEMENTED
SCN-APP-002	res://scenes/vertical_slice/vertical_slice_app.tscn	Hosts workshop/result screens	IMPLEMENTED
SCN-WORK-003	res://scenes/vertical_slice/screens/vs_workshop_screen.tscn	Item state → enhancement/repair/result	IMPLEMENTED / CURRENT_PARITY_REVIEW_REQUIRED
SCN-RESULT-004	res://scenes/vertical_slice/screens/vs_customer_result_screen.tscn	Resolved result → next action	IMPLEMENTED / UX_NOT_RUN
SCN-PREC-005	Not yet approved	+9→+10 entry → outcome	NOT_IMPLEMENTED
SCN-CHR-006	Not yet approved	UID detail → meaningful history	NOT_IMPLEMENTED

17. SCRIPT RESPONSIBILITY MAP

Script / class	Public responsibility	Do not own
VSEnhancementResolver	Resolve a legal attempt from item/target/rolls.	UI copy, hidden manual probability.
VSEnhancementActionService	Commit a resolved attempt, ledger, save boundary.	Drawing or player-facing styling.
VSRepairResolver	Quote/apply repair and structural consequence.	Store price tables beyond approved model.
VSItem	Item UID and durable state mutations with invariants.	Customer narrative or UI.
VSSaveService	Atomic persistence/load/migration boundary.	Rerolling resolved event outcomes.
VSCustomerWorldEventResolver	Actual-use event/damage eligibility and outcome packet.	Direct combat simulation or universal Tag bonus.
VSWorkshopScreen	Display current item and dispatch actions.	Durable numeric authority.
VSApp	Screen orchestration and app state transition.	Rule calculation.

18. SIGNAL AND EVENT FLOW

EVT ID	Emitter → receiver	Payload / timing	Required invariant
EVT-ENH-001	UI → enhancement service/resolver	selected UID, next target, deterministic/live rolls	Disable duplicate input until resolution.
EVT-ENH-002	resolver → UI/save	one of SUCCESS, FAILED_HOLD, FAILED_DAMAGE	Exactly one result and no downgrade/critical.
EVT-REP-001	UI → maintenance/repair resolver	UID, resources, quality/scar rolls	Job consumed exactly once at start.
EVT-USE-001	customer result flow → actual-use resolver	event ID, UID, actual-use profile, damage roll	Purchase/handoff alone rejected; UID result only once.
EVT-SAVE-001	service → save service	candidate V3 envelope	Temp-flush-rename/backup; resolved results do not reroll.
EVT-UI-001	app → hosted screens	state transition	One active surface; return preserves selected UID context.

19. STATE MACHINES

Item life state

BIRTH → ACTIVE → HANDOFF → ACTUAL_USE_RESULT → ACTIVE
 ↘ DAMAGE → REPAIR_ELIGIBLE → ACTIVE
ACTIVE / HANDOFF / RESULT → DESTROYED → ARCHIVED
ARCHIVED → optional NEW_UID_SUCCESSOR (no power/history transfer)

Enhancement attempt state

PREVIEW → CONFIRM → RESOLVE → SUCCESS(+1) | FAILED_HOLD | FAILED_DAMAGE
FAILED_DAMAGE → derived durability state → optional one repair job

Precision state (future)

ELIGIBLE_AT_PLUS9 → READ_LINEAGE_AND_METHOD → PREVIEW → CONFIRM
→ SUCCESS_AT_PLUS10: one Tag resolve/write | failure: no Tag

The precision state machine is DOCUMENTED / CONFIRMED at rule level but NOT_IMPLEMENTED at UI/data-content level.

20. SAVE/LOAD CONTRACT

Save/load must preserve UID, birth facts, level, exact durability fields, meaningful ledger, resolved events, repair-job state, ownership and content-result references. Load must not reroll a resolved attempt, repair quality, scar, or world event. V2 save migration is allowed on read and writes V3 on next save; V1 fails closed. Future DAT-PREC-001 needs an explicit additive migration and backfill policy before any production Tag write exists. Never infer or synthesize a Tag from legacy method history.

21. IMPLEMENTATION TRACEABILITY

Player need	System / content	UI / asset	Code/data/test evidence	State
Stop or push with clear risk	SYS-ENH-001	UI-WORK-001	VSEnhancementResolver, action service, test_vs_enhancement_resolution.gd	IMPLEMENTED / CURRENT_PARITY_PARTIAL
Understand damage/repair	SYS-DUR-003, SYS-REP-004	UI-WORK-001, AST-DUR-002	VSItem, VSRepairResolver, workshop screen, durability/repair GUT suites	IMPLEMENTED / UX_NOT_RUN
First work enters core loop	UX-ONB-001	UI-MAIN-001, UI-FORGE-001, AST-FORGE-003	VSMainMenu, VSFirstForgeCompletionService, first-forge tests	IMPLEMENTED / UX_NOT_RUN
Same UID returns from the world	SYS-CUS-006, CNT-ADVENTURER-01	UI-RESULT-001	content result JSON, event resolver/action service, result-screen tests	PARTIAL
One +10 Tag reflects player choice	SYS-PRE-002	UI-PREC-001, AST-PREC-006	Decision34 and unified contract only	CONFIRMED / NOT_IMPLEMENTED
Player reads meaningful history	SYS-CHR-007	UI-CHR-001, AST-CHR-007	Decision27 and ledger concepts	CONFIRMED / NOT_IMPLEMENTED

22. TEST AND QA CONTRACT

QA ID	Verification	Current result
QA-CAN-001	Python current-authority and core-simplification contracts	AUTOMATED_TEST_PASS in the snapshot's recent document PR checks.
QA-VS-002	GUT vertical slice domain/resolver/screen suites	Existing test files are present; exact full local rerun is required for any code change.
QA-SAVE-003	Save migration, same UID, deterministic resolved events, backup recovery	Contract documented in preset; test evidence must be cited per exact run.
QA-PREC-004	New Tag combination matrix, failure/no-write, Grade/Event isolation, empty-lineage policy	NOT_IMPLEMENTED / BLOCKED_BY_DECISION.
QA-UX-005	Five-minute comprehension, +10 identity recognition, +11 STOP/PUSH understanding, returned-result causality	NOT_RUN.
QA-AND-006	Android portrait safe area, 48dp targets, contrast/large text, rotation, device load	NOT_RUN.
QA-REL-007	Rights/terms, AAB/signing, content rating and Google Play evidence	NOT_RUN / RELEASE_BLOCKED_UNVERIFIED.

Required manual test scenarios

1. New Game → First Forge → Workshop runs without losing the selected UID.
2. At +9, the player can read what lineage and method mean before confirmation; at success exactly one valid Tag is written; at failure none is written.
3. At +11, preview communicates success, hold, and damage outcome in mutually exclusive terms; player can stop without penalty.
4. Verify 5/5/5, 4/4/5, 2/2/5, 1/1/5 presentation and actual modifiers.
5. Actual customer use only, not handoff, can produce one damage event; result has the same UID, 2-4 reasons, and one next action.
6. Kill/restart/load cannot duplicate repair jobs or reroll resolved outcomes.
7. On target Android devices, check 360×640 and 720×1280 logical layouts, font scaling, contrast, one-hand reach, and tap recovery.

23. VERTICAL SLICE DEFINITION

Active playable slice: Slice B. Make one UID weapon; progress ordinary +0→+10; experience craft-rise beats at each five levels; use one +9→+10 precision choice once its content/UI is approved; choose STOP or PUSH at +11→+15; hand off; receive one brief actual-use result; use it to decide repair/continue/new work.

Hypothesis	Risk	Evidence required
Fun	Does +11 make stopping feel like agency, not avoidance?	5+ human think-aloud tests, choice distribution and reasons.
Identity	Does one +10 Tag make the work memorable without turning it into a generic loot system?	Comprehension test and Tag recall after result return.
Production	Can customer return content be produced with 2-4 causal reasons and one next action without bespoke system drift?	Nadia slice plus two content rows using the same template.
Technical	Can V3 save preserve results and repair idempotently across restart?	Automated and Godot runtime save/load evidence.
Visual	Can native UI remain legible over approved workshop art at actual portrait gameplay size?	Client capture plus Android/accessibility/human visual QA.

24. RISKS AND BLOCKERS

Risk ID	Risk	Class	Disposition / next validation
RSK-001	Tag matrix and no-lineage behavior unresolved; any placeholder would create false product meaning.	Product	BLOCK: decide exact data matrix before UI/write implementation.
RSK-002	V2 runtime may drift from current durability/failure/precision canon despite similar field names.	Technical/canon	TEST: contract-to-code audit at exact head before implementation.
RSK-003	Customer life becomes a feature list rather than returned consequence.	Fun/content	TEST: playtest causal recall and next-action comprehension.
RSK-004	Repair economy is accepted test budget, not player-validated price.	Economy	TEST_IN_PLAY: sensitivity then human play.
RSK-005	Runtime asset binding exists but visual usability is unverified.	UX/visual	TEST: client capture, Android, accessibility/human QA.
RSK-006	AI asset commercial/distribution terms and platform profile are incomplete.	Release/rights	MITIGATE: current terms and per-asset rights evidence before shipping.
RSK-007	PR #196 may be mistaken for accepted implementation.	Process	MONITOR: retain read-only, exclude from current architecture claims.
RSK-008	Mobile production scope expands into shop tycoon, direct combat, or dense affix inventory.	Scope	REJECT: preserve enhancement-first, customer-causality support role.

25. USER DECISION REQUIRED

Decision	Why needed	Minimum options / recommended boundary
DEC-PREC-35	Define first catalyst lineage × precision method × Tag content matrix.	Use one stored composite Tag; lineage supplies family, method supplies expression. No fourth slot.
DEC-PREC-36	Define empty catalyst-lineage behavior.	Explicit block, explicit default, or explicit seed flow; do not infer from old records.
DEC-UI-37	Approve UI-PREC-001 entry/exit and required preview copy.	Keep screen as a short choice/confirm/result surface within Workshop flow.
DEC-UX-38	Define Chronicle screen entry and readable event grouping.	Meaningful event timeline only; no routine attempt spam.
DEC-PLAY-39	Approve first human Slice B playtest protocol.	5+ targeted testers; measure understanding, choice reasons, recall, and frustration.

26. IMPLEMENTATION QUEUE

1. Resolve DEC-PREC-35 and DEC-PREC-36; update one current JSON/contract owner and add RED tests.
2. Implement DAT-PREC-001 and resolver/write isolation in the approved current-canon vertical-slice path; run GUT/Python and exact save migration checks.
3. Implement UI-PREC-001 only after flow/copy/accessibility contract; add actual consumer record before any new precision raster.
4. Audit V2 enhancement/durability/repair paths against Decisions28/29/31/32; correct only approved drift.
5. Deliver Nadia actual-use result to runtime with same-UID and causal reason validation; then author two reusable content rows.
6. Execute human Slice B playtest and Android/UI visual QA; tune only after evidence.
7. Complete release rights/platform profile, AAB signing/export and store-quality gates last.

27. CHANGE LOG

Date	Change	Scope
2026-08-28	Created source-snapshot AI production specification paired with the user PDF.	Documentation only; source main b4a3b01.
2026-08-28	Records Decision34 supersession: Tag is resolved by catalyst lineage plus precision method.	Current canon; no runtime Tag write/UI added.
2026-08-28	Records approved runtime asset families and their evidence ceiling.	No new image generation or Notion output.

Appendix A. Benchmark disposition

Reference	Observation	ADOPT / ADAPT / REJECT	Blacksmith application and validation
Diablo IV Tempering/Masterworking	Official update separates explicit selected affix customization from a quality/masterwork path.	ADAPT	One explicit Tag result at +10, separated from ordinary enhancement. Reject its broad loot-scale, capstone randomness, and numeric economy. Validate Tag comprehension and no Grade/Event contamination.
Last Epoch Forging Potential	Crafting boundary is a distinct item-level resource in official support information.	REJECT as a new resource	Blacksmith already owns durability/repair risk; adding forging potential would duplicate the decision. Keep only the principle that the consequence boundary must be readable.
Shop Titans	Mobile craft/sell/shop expansion supports a broad tycoon loop.	REJECT broad tycoon/guild/market loop; ADAPT clear visitor-purpose framing	Do not add multiplayer market, guild, or expanding blueprint treadmill to MVP. Validate that customer purpose increases willingness to make a risk decision.
Moonlighter	Shopkeeping, careful price setting, crafting/enchanting and villagers create a repeated day-life identity.	ADAPT item-to-person return loop	Use same UID return and customer context, not dungeon combat or price-haggle simulation. Validate remembered causal result after one session.
Potion Permit	Diagnosis → gather → make → cure gives each resident a legible need/result chain.	ADAPT	Customer asks and returned result should expose a small causal chain, not a hidden total score. Validate players can name a cause and next action.
Android/Godot official guidance	AAB/signing/export, stability/ANR/memory/bitmap monitoring and quality are distinct release requirements.	ADOPT release gate	Preserve NOT_RUN until actual Android export/device telemetry exists; no release claim from headless tests.

Appendix B. Non-goals

- Direct player combat, dungeon exploration, army/guild micromanagement, betting, a general shop-tycoon market, player keyword/title from +10, generic affix A/B slots, and prestige/reset after +100.
- Any automatic current image generation for a planning/document gap in this two-artifact work package.
- Treating existing automation, an image file, or a Notion record as human UX, runtime visual, Android, or release evidence.